

ALEXANDER CONTRERAS

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EDUCATION

University of California, Irvine

Irvine, CA

M.S. Aerospace Engineering - Fluid Dynamics & Propulsion - GPA: 3.47

December 2026

California State University, Long Beach

Long Beach, CA

B.S. Aerospace Engineering - GPA: 3.56

May 2024

PUBLICATIONS

- "[Polybutadiene structural effects on dynamic wetting](#)" - co-author, *npj Advanced Manufacturing* (2024)

SKILLS

Software: SolidWorks (CAD), MATLAB, Python, Linux, Excel

Engineering: Mechanical Design, Fluid/Thermal Analysis, Integration & Testing, Quality Control, System V&V

EXPERIENCE

University of California, Irvine

Irvine, CA

Graduate Researcher | High-Temperature Flow Modeling

January 2026 - Present

- Developed a repeatable simulation and post-processing workflow for propulsion-relevant non-equilibrium post-shock analysis, enabling consistent cross-case engineering comparisons.
- Automated parameter-sweep case generation, batch execution, and output parsing on Linux using Python, reducing end-to-end turnaround by **60%** across **50+** cases.
- Improved analysis reliability by debugging run failures and standardizing post-processing metrics, reducing rework by 25% and maintaining <2% variation across re-runs.

University of California, Irvine

Irvine, CA

Project | Thermochemical Nonequilibrium Shock Modeling

December 2025

- Developed a 1-D nitrogen shock-layer model in MATLAB/Python using Rankine-Hugoniot jump conditions and a two-temperature N_2/N formulation to analyze post-shock relaxation.
- Implemented and documented a coupled ODE solver for species, velocity, translational temperature, and vibrational temperature; post-processed outputs into engineering plots and equilibrium metrics.

California State University, Long Beach

Long Beach, CA

Project Manager & Thermal Systems Lead | Lunar Rover Design

August 2023 - May 2024

- Designed radiator system in SolidWorks, increasing heat-rejection area to $\sim 22 \text{ ft}^2$ and reducing estimated heat-flux density by **$\sim 59\%$** .
- Developed thermal-control requirements and thermal balance/vacuum test plans for electronics, sensors, and water storage across **-205°C to 100°C** operating limits.
- Documented and managed subsystem requirements, interfaces, and V&V plans to support rover integration, design reviews, and test readiness.
- Conducted subsystem trade studies balancing mass, power, cost, and integration constraints, reducing estimated project cost by **35%**.

California State University, Long Beach
NSF Researcher | Space & Combustion Lab

Long Beach, CA
June 2021 - May 2024

- Conducted benchtop experiments to evaluate binder wettability in solid rocket system applications.
- Automated MATLAB data reduction for **100+** high-speed test videos, extracting time-resolved metrics, and generating standardized plots and tables, reducing processing time by **50%**.
- Performed hardware inspections, followed standardized test procedures, and maintained quality control records to improve repeatability.
- Reported weekly test results and resolved experimental and analysis issues to improve procedure robustness.

University of California, Berkeley
Inclusive Excellence Researcher | Soft Matter & Interfaces

Berkeley, CA
June 2022 - July 2022

- Processed microscopy data in MATLAB to quantify droplet size, volume fraction, and water transport across oil/surfactant trials.
- Designed microfluidic device concepts in SolidWorks to improve flow control, leakage resistance, and droplet repeatability.
- Conducted water-in-oil droplet experiments to evaluate interface stability, surfactant effects, and transport behavior..
- Presented research results to 50+ attendees at the Berkeley 2022 Summer Symposium.

LEADERSHIP & ACTIVITIES

Professional Tutors of America Inc.
Tutor

Orange County, CA
June 2023 - Present

- Tutored students across multiple grade levels in math, physics, and chemistry, translating technical concepts into clear, approachable explanations for different learning needs.
- Developed schedules, practice materials, and academic follow-ups to keep students organized, accountable, and prepared for coursework.
- Coordinated with teachers, tutors, and program staff to communicate student progress and support an inclusive, team-oriented learning environment.

MathWorks
MATLAB Ambassador | CSU Long Beach

Long Beach, CA
October 2023 - May 2024

- Led workshops introducing students to MATLAB and Simulink for design, modeling, analysis, and simulation applications.
- Developed supplemental learning materials to help students apply MathWorks tools to coursework and projects.